

HASHIN GAYASHA MAHARAMBAGE

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PROFESSIONAL SUMMARY

Fourth-year Mechatronics Engineering undergraduate at Curtin University Colombo maintaining an 80.52% course weighted average. Hands-on experience in industrial automation, robotics, embedded systems, and mechanical design, demonstrated through two completed engineering internships at Ansell Lanka and C3 Labs. Proven track record in PLC programming (Omron, Siemens), ROS 2 navigation, VHDL/FPGA design, cobot deployment, and leadership within student governance.

EDUCATION

Bachelor of Engineering (Hons) in Mechatronics Engineering

Curtin University Colombo, Sri Lanka

Jul 2023 – Jun 2027

Status: Fourth Year Undergraduate | Course Weighted Average (CWA): 80.52%

G.C.E. Advanced Level & Ordinary Level

Thurstan College, Colombo

Jan 2009 – Jan 2022

G.C.E. Advanced Level: Combined Mathematics (B), Physics (C), Chemistry (C)

G.C.E. Ordinary Level: 8 Distinctions (A) including Mathematics, Science, History, English, Sinhala, Commerce, English Literature, and Information Technology

ENGINEERING EXPERIENCE

Engineering Intern

Ansell Lanka

Nov 2025 – Feb 2026

- Designed and commissioned the BPE Folding Machine from scratch utilizing an Omron CP1E PLC (CX Programmer), soft-start pneumatics, 20 reed switches across 10 actuators, RS-232 serial communication, and a Kinco HMI with Auto, Manual, and E-Stop operational modes.
- Programmed a DOBOT CR12A cobot via teaching pendant for pick-and-place tray operations; configured PLC I/O signals and holding registers for bidirectional real-time data exchange.
- Reverse-engineered a Ven Pack machine frame and linear actuator setup, producing SolidWorks engineering CAD models and performing lead-screw drive calculations for motor selection.
- Optimized operational safety and reliability by troubleshooting PLC-based control circuits and refining pneumatic mechanical assemblies.

Research & Development Intern

C3 Labs

Nov 2024 – Feb 2025

- Developed embedded C++ firmware for e-paper display integration on IoT Air Quality Monitoring (AQM) devices and executed routine calibration maintenance for sensor accuracy.
- Implemented a BLE-based self-payment system featuring secure QR code processing using Flutter for offline financial transactions.
- Collaborated within cross-functional teams under tight deadlines to deliver high-priority R&D deliverables.

TECHNICAL PROJECTS

Advanced Digital Design — Automated Crazy Machine

Curtin University Colombo

2026

- Engineered an autonomous electromechanical system (40×40×40 cm) transporting a steel ball through a 9-stage sequence managed by an Altera DE1 FPGA running a custom VHDL Finite State Machine (FSM).
- Integrated NEMA 17 stepper motors with A4988 1/16 micro stepping, servo motors, electromagnets, and optical/LDR sensor feedback. Executed 3D mechanical CAD design in SolidWorks and rapid 3D printing prototyping.

Karto SLAM & Autonomous Navigation

Curtin University Colombo

2026

- Implemented real-time 2D mapping and graph-based pose optimization using the ADLINK slam_karto_ros2 open-source package within ROS 2.
- Simulated a TurtleBot4 mobile robot in Gazebo to analyse scan matching, pose-graph generation, loop closure, and data association integrated with Nav2 for path planning.

ROS 2 Robot Manipulation & Digital Twin

- Curtin University Colombo2026
- Developed a ROS 2–based teleoperation system for the DOBOT Magician robotic arm enabling joint-space and Cartesian-space keyboard control.
 - Programmed autonomous block stacking subroutines and established a synchronized RViz digital twin for real-time robotic state visualization.

Microcontroller Project — Dual-Mode Robotic Car

- Curtin University Colombo2025
- Constructed an ESP32-CAM autonomous maze-navigating vehicle with secondary HC-05 Bluetooth manual control using an Arduino Mega running C code.

Line-Following Cart & GUI

- Curtin University Colombo2025
- Designed an Arduino Nano Bang-Bang speed controller with an analog PWM signal chain (DAC + LM741 op-amp + LM339 comparator + 555 timer) paired with a real-time Python GUI for sensor telemetry.

Chemical Treatment Process SCADA Simulation

- Personal Project2024
- Developed Siemens S7-400 ladder logic control routines for industrial tank filling, fluid heating, and pump sequencing with a WinCC Flexible SCADA interface.

3-Level Elevator Automation Model

- Personal Project2023 – 2024
- Programmed a PLC ladder logic algorithm for a 3-floor selection elevator with priority call handling.

QUALIFICATIONS, CERTIFICATIONS & LEADERSHIP

PLC Programming Certification

- SLIR Institute
- Completed comprehensive industrial automation training covering Siemens, Mitsubishi, and Xinje PLC platforms, HMI/SCADA integration, motor control circuits, and industrial troubleshooting.

Editor

- IEEE Student Branch of Curtin University Colombo2026
- Technical content review, and article editing for the IEEE student branch community.

Assistant Treasurer

- Curtin Colombo Student Guild2024 – 2025
- Managed financial operations, budgeting, and resource allocation for student activities at the university level.

President

- Photographic Society of Thurstan College2020 – 2021
- Led society operations, event organization, team coordination, and workshop execution.

TECHNICAL & PROFESSIONAL SKILLS

Robotics & Automation: ROS 2, Gazebo, PLC Programming (Siemens S7-400, Omron CP1E, Ladder Logic), Kinco HMI, WinCC SCADA, DOBOT CR12A / Magician

Mechanical & Hardware CAD: SolidWorks, AutoCAD, Pneumatics Circuit Design, Actuator Selection & Mechanical Calculations

Embedded Systems & Electronics: VHDL / FPGA (Quartus, Altera DE1), Arduino, ESP32, Raspberry Pi, Microcontrollers

Programming Languages & Tools: C, C++, Python, MATLAB / Simulink, Git

Professional & Soft Skills: Engineering Ethics, Team Leadership, Technical Documentation, Project Management, Strategic Problem-Solving

ADDITIONAL INFORMATION

Languages: English (Professional / Fluent), Sinhala (Native)

Interests: Autonomous Robotics, AI/ML in Perception Systems, Product Design, Photography

References: Available upon request